BusTopia Layer Architecture

B1_04

Layers

- Presentation Layer
- Business Layer
- Persistence Layer
- Database Layer
- Integration Layer
- Infrastructure Layer

Presentation Layer

Purpose

- Handle user interaction by providing user interface and communicating with backend services.

Presentation Layer

Components/Services

- User Interface

Displays Components for general users:

- User profile: Displaying user profile
- Trip History: Displays all the previous trips of the user
- User Reviews: Displays reviews of the user.
- User Complaints: Displays complaints of the user.
- Bus information : Display bus information
- Ticket management: Displays seats, available tickets, buy tickets, find previous tickets, and verify tickets.
- Smart route information: Takes source and destination and displays smart routes.
- Bus Service Review: Display and add review for a bus service
- Live tracking: Displays current live location of a bus.

- Technology: Next.js for single page applications , Tailwind CSS for styling

Presentation Layer

Components/Services

- User Interface

Displays Components for Admin:

- Admin profile: Displaying admin profile
- Bus Service Complaints: Displays complaints and takes feedback.
- Notification: Displays notifications.
- Live tracking: Displays current live location of a bus.
- Schedule and Route Analysis: Displays analysis data and suggestions.
- Review Analysis: Displays analysis of the review for a bus service provided by the users.

- Technology: Next.js for single page applications , Tailwind CSS for styling

Presentation Layer

Components/Services

- API Gateway
  - Acts as an entry point to connect frontend to backend services
  - Technology: Next.js own API

Business Layer

Purpose

- Coordinate workflows, implements the core logic of the system.

Business Layer

Components/Services

- Ticket Service: Managed ticket related logic
- Smart route detection service: manage smart route-finding logic
- Payment Service: Manage payment validation and coordination with the integration layer.
- Review Service: Manage review process and set new reviews

Business Layer

Components/Services

- Notification Services: Manages notification services
- Complain and feedback: Manages complain and feedback
- Live tracking Services: Manages the live tracking logic of the buses.
- User behavior analysis: Manages the logic for analyzing the information of the ticket and bus demand of different times.

- Technology: Next.js, Node.js

Persistence Layer

Purpose

- Abstract information with the database.

Persistence Layer

Components/Services

- Data Access Objects (DAO): Encapsulates database operations for different database entities
  - User information DAO
  - Bus Information DAO
  - Seat and Ticket information DAO
  - Schedule and route information DAO
  - Review DAO
  - Complaints and feedback information DAO
  - Notification DAO
- Technology: Node.js to query data, salination to validate input data

Persistence Layer

Components/Services

- Validation: Ensures the provided input meets the correct format which ensures consistent business environment
- Technology: Joi(Node.js) Validator.

Database Layer

Purpose

- Handle the actual storage and management of data in a relational database with different entities to represent different types of data.

Database Layer

Components/Services

- Relational databases: Stores structured data with tables for separate entities.
  - BusService
  - BusTicket
  - BusReview
  - BusSchedule
  - BusRoute
  - User
  - BusComplaint
  - BusStop
  - Notifications
- Technology: PostgreSQL

Integration Layer

Purpose

- Facilitates communication between external services and the APIs we are using.

Integration Layer

Components/Services

- Payment Gateway: Manages interaction with different payment methods.
- Technology: SSLCommerz.

Integration Layer

Components/Services

- GPS tracking: Get real time bus locations (Coordinates) for live tracking feature
- Technology: Opencage Geocoder API.

Integration Layer

Components/Services

- Traffic APIs: External APIs will calculate the estimated time to reach a certain destination or to find the efficiency of a specific path.
- Technology: Google Maps API

Infrastructure Layer

Purpose

- Provide functional support for hosting, run, deployment and to maintain the project.

Infrastructure Layer

Components/Services

- Hosting platform: Automates and Runs the application on a scalable cloud infrastructure.
- Technology: AWS

Infrastructure Layer

Components/Services

- Containerization: Ensures consistent deployment across different environments.
- Technology: Docker

Infrastructure Layer

Components/Services

- CI/CD pipeline: Automates the build tests and the deployment process.
- Technology: Jenkins

Thank you